Amish Sethi

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Research Interests

Reliable embodied AI, foundation models that connect video, language, and robot action, scalable neurosymbolic learning, and understanding and adapting large generative models.

Education

Harvard University Cambridge, MA

Ph.D. in Computer Science, incoming *Fall 2026*

Advised by Heng Yang and Yilun Du. Kempner Institute Graduate Fellow and NSF Graduate Research Fellow.

University of Pennsylvania Philadelphia, PA

B.S.E. in Computer Science and M.S.E. in Computer and Information Science *May 2026*

Cumulative GPA 4.00 out of 4.00.

Honors and Awards

- NSF Graduate Research Fellowship, National Science Foundation *2026*
- Kempner Institute Graduate Fellowship, Harvard University *2026*
- John Grist Brainerd Award, Penn Engineering, top graduating senior *2026*
- CRA Outstanding Undergraduate Researcher Award, Honorable Mention *2025*
- NeurIPS 2025 Spotlight for ESCA, top 3% of submissions *2025*
- First Place, International Public Policy Forum *2022*
- National Merit Scholar; ISEF Finalist in genetics research *2021*

Publications

My name is shown in bold. Authors marked with an asterisk contributed equally.

Jiani Huang*, Brandon Y. Yang*, **Amish Sethi***, Yuchen Zheng, Christopher Watson, Jianing Qian, Mayur Naik, Dinesh Jayaraman. "Retrieval-Augmented Vision-Language-Action Model." Under review at *CoRL 2026*. Senior thesis.

Amish Sethi, Boya Zeng, Wenhao Chai, Zhuang Liu. "Do Diffusion Models Learn to Generalize Basic Visual Skills." Under review at *NeurIPS*.

Zhiqiu Xu*, **Amish Sethi***, Mayur Naik, Ser-Nam Lim. "Delta Activations: A Representation for Finetuned Large Language Models." Under review at *COLM*; *NeurIPS 2025 ER Workshop*. [arXiv]

Jiani Huang, **Amish Sethi***, Matthew Kuo*, Mayank Keoliya, Neelay Velingker, JungHo Jung, Ziyang Li, Ser-Nam Lim, Mayur Naik. "ESCA: Contextualizing Embodied Agents via Scene-Graph Generation." *NeurIPS 2025*, Spotlight, top 3%. [Website]

Aaditya Naik, Jason Liu, Claire Wang, **Amish Sethi**, Saikat Dutta, Mayur Naik, Eric Wong. "Dolphin: A Programmable Framework for Scalable Neurosymbolic Learning." *ICML 2025*. [arXiv]

Neelay Velingker, **Amish Sethi***, Jason Liu*, William Dodds*, Zhiqiu Xu, Saikat Dutta, Mayur Naik, Eric Wong. "CLAM: Unifying Finetuning, Quantization, and Pruning by Chaining LLM Adapter Modules." *ICML 2024 ES-FoMo Workshop*. [Paper]

Andrew Ni*, **Amish Sethi***. "Functional Genetic Biomarkers of Alzheimer's Disease and Gene Expression from Peripheral Blood." *bioRxiv*, 2021. [Paper]

Research Experience

University of Pennsylvania Philadelphia, PA
Research Assistant, advised by Mayur Naik and Dinesh Jayaraman *2023 - 2026*

- Built RA-VLA, a retrieval-augmented vision language action policy, and VINE, the scene-graph model behind ESCA, improving manipulation and cutting perception errors with no new data.
- Led the experimental validation for Delta Activations and released more than 700 finetuned models, and contributed to scalable neurosymbolic learning (Dolphin) and unified LLM compression (CLAM).

Princeton University Princeton, NJ
Research Intern, advised by Zhuang Liu *2025*

- Designed a controlled framework that isolates basic visual skills to probe diffusion model generalization, finding strong interpolation but weak extrapolation.

Open-Source and Datasets

- **VINE**, a promptable model that turns video into spatio-temporal scene graphs, released with the ESCA-Video-87K dataset on Hugging Face.
- Released more than 700 finetuned open-source models on Hugging Face for Delta Activations. **LASER**, the codebase behind VINE, passed 100 stars on GitHub and was featured on the Google Open Source Blog.

Experience

Research Lead San Francisco, CA
AfterQuery *May 2026 - Present*

- Leading research on the next large benchmark for embodied agents and on scaling robotics data collection beyond teleoperation, including a more capable handheld interface for in the wild demonstrations.

Research Intern Singapore
National University of Singapore *2023*

- Built FIIGNET, a generative pipeline that synthesizes diseased-fish images for aquaponics early detection, improving detection accuracy by 17%.

Computer Vision Intern Pittsburgh, PA
RoadBotics *2021*

- Built a Mask R-CNN traffic-sign detector at 90% accuracy from video, deployed by the Pennsylvania state government for road-asset inventory.

Teaching and Service

Head Teaching Assistant, CIS 7000 Large Language Models, Penn *Fall 2024*

- Planned curriculum, designed assignments, and lectured on efficient finetuning and evaluation for Penn's first dedicated LLM course of more than 120 students.

Undergraduate Research Mentor, Penn PURM *Summer 2024*

- Mentored five undergraduates on the CLAM project in research methodology and scalable optimization.

Service. Reviewer for the ICML 2024 ES-FoMo Workshop, AAAI 2026, and ICLR 2026, and recipient of an 8,000 dollar Penn Grant for Faculty Mentoring Undergraduate Research for work on neurosymbolic AI.